

LINEAR ALGEBRA FOR COMPUTING

MAT 2400 Spring 2025

COURSE INFORMATION

SEC 1: MWF 10:40-11:30am Mendel 258
and T 10:30-11:20am Old Falvey 101
MWF 12:50-1:30pm in Mendel G86
SEC 2: and T 11:30am-12:20pm Old Falvey 101

[Link to Campus Map](#)

TEXT: There is no required textbook for this course. See Blackboard for links to recommended texts.

PROF: Nathan Corwin

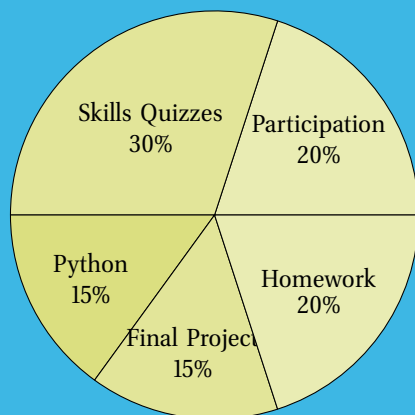
OFFICE M 3:30-4:30pm,

HOURS: Tu 1-2:30pm,
Th noon-1:30pm
SAC 472

nathan.corwin@villanova.edu

GRADES

A	A-	B+	B	B-	C+
93	90	87	83	80	77
C	C-	D+	D	D-	F
73	70	67	63	60	<60



Values above are minimum score to guarantee the letter grade.

All course documents, including this one, can be found on Blackboard.

*Learning to solve problems is education of the will. Solving problems which are not too easy for them, students learn to persevere through failure, to appreciate small advances, to wait for the essential idea, to concentrate with all their might when it appears. **If a student had no opportunities in school to become familiar with the varying emotions of the struggle for the solution, then their mathematical education failed in the most vital point.*** –Paraphrased from *How to Solve It*, by George Pólya (1945)

SKILLS AND PRACTICES

Apply linear algebra notions to real-world problems;

Recognize and interpret situations in which vectors or matrices represent data;

Make connections between computational tasks in linear algebra, theoretical concepts, and application concepts;

Independently research a linear algebra technique and application and explain it in a technical report;

Make use of the Python programming language to carry out linear algebra computations.

TYPES OF ASSESSMENTS

Homework: Homework will be assigned approximately weekly. Homework should be uploaded to blackboard as a **single pdf**. I will not be responsible for converting jpegs and iPhone images into a pdf so that I may grade your assignment. Late written homework is not accepted except for extenuating circumstances. You must email as soon as possible if you encounter such circumstances in order to receive credit for the assignment. Working together on homework is encouraged, however you must list the names of those you work with on each written assignment. The lowest homework grade will be dropped.

Participation Assignment: This course will require you to take an active role in your learning. Interspersed with lectures you will be asked to watch videos outside of class and answer questions about that as well as do in class activities that include work by hand as well as in Python. While attendance will not be formally taken you will be responsible for completing any in-class activities that you miss due to absences. It is recommended that you attend class however since we will not be following a single textbook and the activities are most beneficial when worked on in the company of your classmates and your instructor. You will be required to turn in a completed after video-questionnaires, activity worksheets, and Jupyter notebooks. These will be graded mostly on effort/completion.

Skills Quizzes: Approximately every other week we will take 15-25 minutes on Monday to complete a skills quiz. These are to ensure you are retaining the most basic concepts and problem solving skills required for linear algebra. They will be short and focus on quick computational type problems.

Python Projects: You will be asked to complete 3 python projects throughout the semester in which you will explore more deeply a concept discussed in class. In these you will also be writing your own python code, or modifying given code to complete certain tasks. The aim of these projects is to deepen your knowledge of python when it comes to linear algebra as well as to see linear algebra in practice in the context of an application. These will require a written report to be submitted along with your python code. The specifics of what goes into each report will be detailed in the project statement.

Final Project: There are many applications/examples in the book and in other sources that we will not have the time to cover in detail. A list will be maintained on Blackboard. For your final project, you may choose one of the ideas listed, or choose a different application that interests you. More details will come later in the semester.

WHAT ARE OFFICE HOURS?

Office Hours are times set aside during the week for professors to meet with students. In this class, you don't need an appointment to come to office hours – just drop by! Here are a few things I recommend for making office hours productive and fun:

- Try the homework before asking me about it. This does not mean you must have (almost) solved it. It means you should have thought about the question before asking me about it. Look up definitions in your notebook or textbook, find a similar example from class, etc.
- If there are other people waiting to see me, say hi, introduce yourself, and see if you have the same questions. You can work together (and I encourage you to work together!) and then I can talk to a bunch of you.
- Know the difference between collaboration (good) and copying someone else's solution (bad). Active collaboration can take many forms: asking "Why?" a lot; explaining your thinking to someone else; passing the chalk around a group of people and taking turns at the blackboard working on a problem.
- Collaboration is one of my favorite parts of being a mathematician: enjoy it! Make friends! Solve problems! Get stuck! Get unstuck!
- Don't be scared to come and ask me questions. I don't think badly about students that come to office hours; if anything it is the opposite. I get to know you better and can understand what is causing you to struggle.

CLASSROOM ENVIRONMENT

The American Mathematical Society (the largest professional society for mathematicians) outlines its vision for a welcoming environment as follows:

The AMS strives to ensure that participants in its activities enjoy a welcoming environment. In all its activities, the AMS seeks to foster an atmosphere that encourages the free expression and exchange of ideas. The AMS supports equality of opportunity and treatment for all participants, regardless of gender, gender identity or expression, race, color, national or ethnic origin, religion or religious belief, age, marital status, sexual orientation, disabilities, veteran status, or immigration status.... A commitment to a welcoming environment is expected of all attendees at AMS activities, including mathematicians, students, guests, staff, contractors and exhibitors, and participants in scientific sessions and social events.

– AMS Policy on a Welcoming Environment

I am committed to the same vision for our classroom environment, and I sincerely thank you for your contributions toward making our classroom (and office hours sessions) a lively and respectful community of thinkers. Please let me know if you feel that we have strayed from this vision at any point during the semester.

YOUR RESPONSIBILITIES

Work Ethic One time estimate for succeeding in an average college course is that 2 or 3 hours of work are required for each hour of class. For this class, that translates to 8-12 hours of work outside of class each week. (If you're spending **more** time than this on Linear Algebra, let's talk!). These hours should include the time spent attempting homework, reading the optional textbooks, and reading over your class notes.

Class Attendance Although I do not formally take attendance, I expect you to attend class. It's okay if you need to miss a class occasionally, but it's your responsibility to make up the material you missed by getting the notes from a classmate and reading the textbook. (By University Policy, Freshman attendance is "mandatory". If a Freshman has more than 6 unexcused absences, I must assign a final grade of "Y" which is a failing grade.) If you are going to miss a class, please let me know as soon as possible; preferably before class.

Reading This course is fast-paced. It will also cover material that may be different than you have seen before. Some of the homework that you do will go above and beyond what is covered in lecture, and the lectures will not cover everything in the textbook. This means that I expect you to read the textbook on your own and fill in details that we don't discuss in class.

Accommodations, Conflicts, & Makeup Exams Except in the case of severe illness or emergency, I need advance notice so that I can schedule a make-up midterm for you. The final exam is scheduled by the registrar and can be rescheduled only in very specific circumstances: mostly for three finals on the same day.

Academic Integrity Policy Don't cheat! Plagiarism in mathematics happens when you look up a solution and use **any part of it** in your writing, even if you could have thought of the ideas on your own! This is one reason that using online resources (Chegg, Slader, etc) is prohibited. (The other reason is that it's bad for your mathematical development.) Quoting the University Academic Integrity "Students who believe that an integrity violation has occurred should report the suspected violation to the faculty member." I expect you to honor these commitments as Villanova community members.

Getting Stuck Being stumped is part of learning mathematics, so please attempt to solve homework problems on your own before asking for help on them. After you've given them a shot, I expect you to seek help when you need it. This can take many forms: asking questions in class; chatting with other students in the class, or other students who took the course previously, about the material; coming to my office hours and working with me and other students. However, let me repeat that finding and copying an answer (online, in a solutions manual, or on someone else's paper) is **plagiarism**, and that's not allowed (it also doesn't count as getting help!).

RESOURCES

MLRC: Provides free drop-in tutoring most afternoons and evenings; individual tutors can be arranged, too. The tutors are advanced students who can help you with homework problems and studying. The MLRC is located in 204 Falvey Library.

Need support in your other classes?
[Academic Support](#)

Counseling Center: Offers individual and group therapy, peer counselors, psychiatric treatment. They also have some Student Discussion Meetings. Call 610-519-4050 or stop in at Room 206 of the Health Services Bldg;
[Counseling Center](#)

Health Center: Offers a variety of medical services for students.
[Health Center](#)

Residential Life: Your RA, your residential area director, and Assistant Director for Housing Services Kelly Dougherty can help you with your living situation.
[Res Life](#)

Student Life: Assistant Director for Student Support Jennifer Liberato can provide support for students experiencing varying degrees of personal distress. She can be contacted at 610-519-4154 or by email at jennifer.liberato@villanova.edu

Chaplain's Office: Provides a range of programming, from religious services and conversations to service and social justice experiences.
[Campus Ministry](#)

International: The International Students Services Office at Villanova University offers a myriad of services to international students.
[Office of International Students](#)

OIA: The Office of Intercultural Affairs creates, maintains, and supports programming and mentoring initiatives that seek to foster and sustain a community of respect and inclusion for Villanova students with a special focus on underrepresented students.
[Intercultural Affairs](#)

National Talk and Text Suicide Lines:
1 (800) 273-TALK or text 741-741. If you want a counselor of color, text "STEVE" to 741-741

HELP!

Depressed? Anxious? Overwhelmed? Sick? Tired?

Talk to me! I'm not a counselor, but I am a human being. Together, we can come up with a plan to use campus resources to help you feel better. Please know that I assume every student is putting forth a good effort in my classes, and more than anything, I want to help you succeed in my class and at Villanova. You can also see a list of resources at the side of the page.

PERSONAL DAYS

Students are entitled to two excused absences for any reason that may contribute to their personal wellness. Students must advise the instructor by email before class of their intent to utilize a Personal Day as the reason for their absence. A Personal Day will not be approved retroactively. Students may, but are not required, to provide additional information regarding their absence. Additionally, Personal Days may not: be used on consecutive class days; be used in the same week; be used immediately preceding or following a University holiday or break period; be used on days when exams, presentations or other major assignments are scheduled.

A Personal Day does not grant an automatic extension for items due. Students remain responsible for all assignments, exams, presentations, etc. due on that date. It is in the instructor's discretion to determine whether any extension is appropriate given individual circumstances.

ETC.

Blackboard will contain weekly overviews of the course. That is also where you will find the weekly homework assignments. I will hopefully remember to also post in-class worksheets there and sometimes solutions.

REQUIRED

Academic Integrity: All students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. You may view the [University's Academic Integrity Policy and Code](#) for a detailed description.

Office of Disabilities and Learning Support Services: It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. All students who need accommodations should go to [Clockwork for Students via myNOVA](#) to complete the Online Intake or to send accommodation letters to professors. Go to the LSS website <http://learningsupportservices.villanova.edu> or the ODS website <https://www1.villanova.edu/universitylife/ods.html> for registration guidelines and instructions. If you have any questions please contact LSS at 610-519-5176 or learning.support.services@villanova.edu, or ODS at 610-519-3209 or ods@villanova.edu

Absences for Religious Holidays:

Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University's obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with their professors as soon as possible, normally at least two weeks in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence.

https://www1.villanova.edu/villanova/provost/resources/student/policies/religious_holidays.html

I reserve the right to make changes to the above class policies at any time.

RIGHTS OF THE LEARNER

As a student in this class, you have the right:

- to be confused,
- to make a mistake and to revise your thinking,
- to speak, listen, and be heard, and
- to enjoy doing mathematics.

WHO IS PROFESSOR CORWIN?

I'm Nathan Corwin. I am an Associate Teaching Professor in the Department of Mathematics and Statistics. I've been at Villanova since 2016. I'm an alumnus of Highland Park High School (1998), the University of Wisconsin (B.S. 2002), the University of Oregon (M.S. 2004 or 2005; sort of both), and the University of Nebraska-Lincoln (Ph.D. 2013). You might notice a little gap in the timeline after my master's degree; that's because I dropped out of graduate the first time around. I worked in politics, taught at three different colleges and universities as an adjunct, and eventually went back to become a professional mathematician.

My research is in a part of algebra called geometric group theory. My focus on the extended family of Thompson groups. These are groups of homeomorphisms of nice topological spaces (like the unit interval or a circle).

There is a connection between my research and computer science. The math side has started to use ideas of complexity theory and automata theory that computer scientists came up with to help understand these groups.

COURSE AI POLICY

This course permits you to use artificial intelligence (AI) tools, such as chatbots, text generators, paraphrasers, summarizers, or solvers, to get guidance on assignments, as long as you do so in an ethical and responsible manner. Essentially, you can think of these tools as ways to help you learn but not to entirely create work for assignments like homework or projects. AI is more like your tutor or TA, not a replacement for your independent thinking. This means that you must:

- Not use AI tools to replace your own thinking or analysis or to avoid engaging with the course content.
- Cite or explain any AI tools you use. Provide the name of the AI tool, the date of access, the URL of the interface, and the specific prompt or query you used to generate the output.
- Be transparent and honest about how you used the AI tool and how it contributed to your assignment. Explain what you learned from the AI tool, how you verified its accuracy and reliability, how you integrated its output with your own work, and how you acknowledged its limitations and biases.
- You are accountable for any mistakes or errors made by the AI tool. Do not rely on the AI tool to produce flawless or correct results. Always check and edit the output before submitting your work.

Using AI tools in an unethical or irresponsible manner, such as copying or paraphrasing the output without citation or transparency, using the output as your own work without verification or integration, or using the output to misrepresent your knowledge or skills, is considered a form of academic dishonesty and will result in a zero grade for the assignment and possible disciplinary action. If you have any questions about what constitutes ethical and responsible use of AI tools, please consult with me (the instructor) before submitting your work.

COVID 19

The potential of a COVID 19 infection affects everyone. To date, the World Health Organization has identified hundreds new strains of coronavirus in America alone, with 20 different variants found in the United States as of December 9, 2024 (8 of which are circulating).

The potential of a COVID 19 infection affects everyone. Villanova University asks every person to share in the responsibility of keeping one another healthy by placing Community First. This is clearly articulated in The CARITAS Commitment statement found: <https://www1.villanova.edu/university/return-to-campus/commitment.html>

Villanova University thrives when ALL of its members are supported and protected.

Villanova University is extremely fortunate that the majority of faculty, students, and staff have been vaccinated. However, there are some members of our immediate community who, due to personal health reasons, cannot undergo the vaccination process at this time or for whom the vaccine will not be effective.

Furthermore, many members of the faculty and staff (including the instructor of this class) have young children at home, some of whom cannot be vaccinated. Some take care of infirmed or elderly family members. The ease of transmission for some of the variants of COVID is especially concerning for these populations.

Our community does not end at the campus property lines. We need to take steps to safeguard ALL people, even those outside of our classroom.

In regards to masks in the classroom, I make two requests. First, if you are not vaccinated please wear a mask in class. Second, anyone who feels any symptoms (fever, runny nose, cough, etc.) is requested to wear a mask.

I thank you in advance for your cooperation.